Ch. 26 HW #1

Principles of Macroeconomics

February 17, 2026

Quick Quiz

1. (b) – The U.S. has grown at roughly 1.70% per year. To figure out roughly how long real GDP would take to double, we can solve

$$1.02^x(\text{GDP}) = 2(\text{GDP})$$
$$1.02^x = 2$$
$$x \log 1.02 = \log 2$$
$$x = \frac{\log 2}{\log 1.02} x = 35$$

2. (c) – U.S. real GDP per capita was roughly \$64,000 while Indian real GDP per capita was roughly \$6,400.
3. (a) – China grew at roughly 2.56% while the U.K. grew at 1.40%
4. (b) – Human capital boosts productivity which in turn boosts real GDP per capita – average real incomes.
5. (d) – See the case study in 26.2.b.

Questions for Review

1. The level of a nation's GDP measures its total income, total expenditure, and total value-added simultaneously. The growth rate of GDP measures how quickly those three quantities are increasing. In the short term, I would rather live in a nation with a high level of GDP because that nation will have a higher average income, and therefore it's likely to have a higher standard of living. In the long term, I would rather live in a nation with a high growth rate because that nation will eventually have a higher average income.
2.
    - Physical capital per worker – the amount of equipment and structures used to produce goods and services relative to the number of workers

- Human capital per worker – the knowledge and skills that workers have
- Technological knowledge – society's understanding of how to best produce goods and services
- Natural resources per worker – the environment assets that an economy has relative to the number of workers

3. A college degree is a form of human capital because it represents a level of knowledge and skill that a worker has obtained.